

Principles of Cryptography

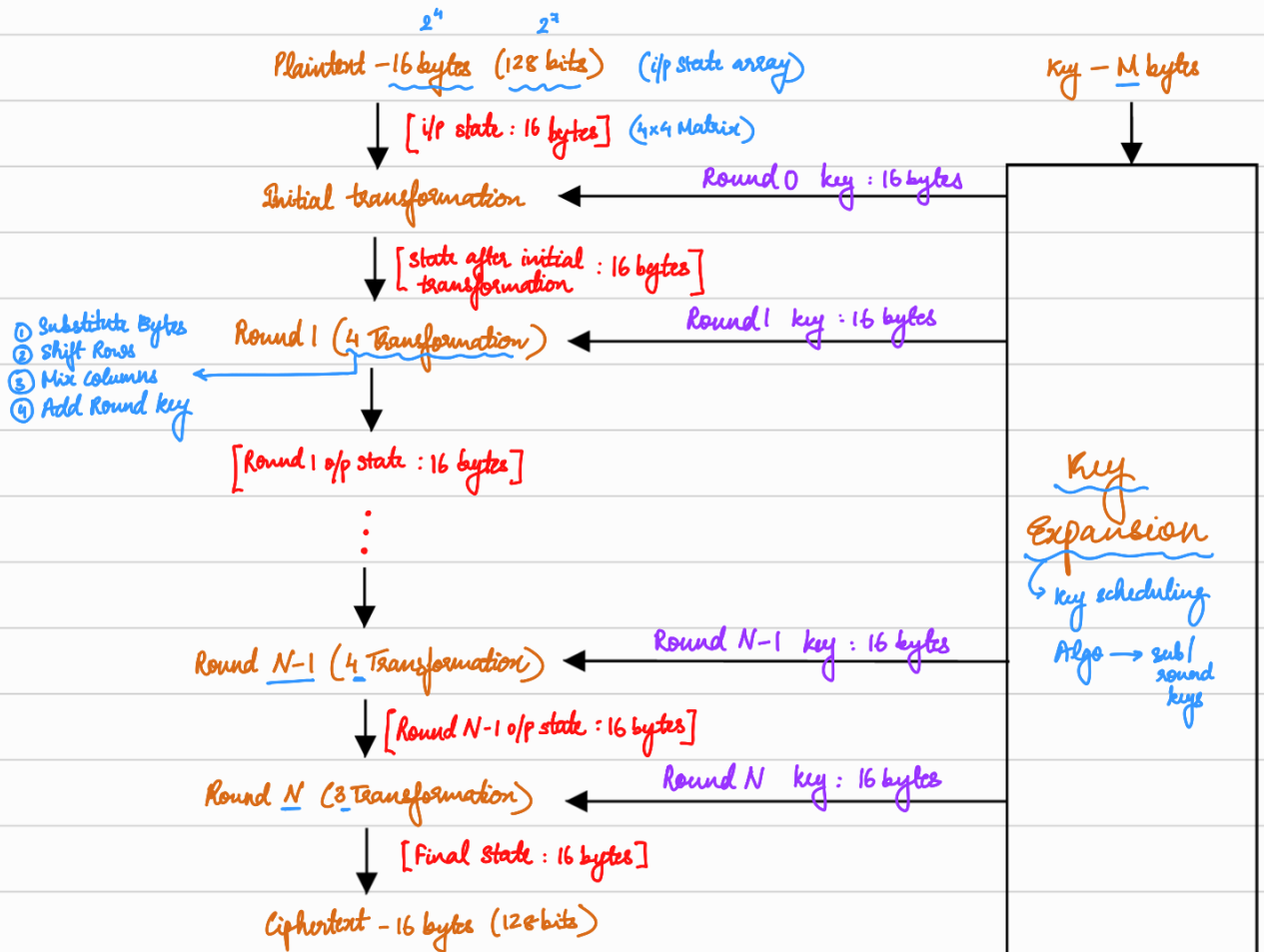
ft. Neso Academy

5/10/25

⇒ AES (Advanced Encryption Standard)

→ It is a symmetric block cipher

→ AES structure



diff. no. of rounds & the key size
rel. b/w $N \times M$

no. of rounds (N)	Key size (in bits) (M)
10 $\left[\begin{array}{l} 0-9^{th} r: 4T \\ 10^{th} r: 3T \end{array} \right]$	128 (AES-128)
12 $\left[\begin{array}{l} 0-11^{th} r: 4T \\ 12^{th} r: 3T \end{array} \right]$	192 (AES-192)
14 $\left[\begin{array}{l} 0-13^{th} r: 4T \\ 14^{th} r: 3T \end{array} \right]$	256 (AES-256)

Numericals (after midsem)

- Q) generate a sequence of random no.'s (show first 6 no.'s) which would be generated using LCG with modulus = 123, seed = 73, multiplier = 5 and increment = 2.

A: $m = 123; X_0 = 73; a = 5; c = 2$

$$X_{n+1} = (a * X_n + c) \bmod m$$

$$n=0; X_{0+1} = (5 \times 73 + 2) \bmod 123$$

$$X_1 = 367 \bmod 123$$

$$X_1 = \underline{\underline{121}}$$

$$n=1; X_{1+1} = (5 \times 121 + 2) \bmod 123$$

$$X_2 = 607 \bmod 123$$

$$X_2 = \underline{\underline{115}}$$

$$n=2; X_{2+1} = (5 \times 115 + 2) \bmod 123$$

$$X_3 = 577 \bmod 123$$

$$X_3 = \underline{\underline{85}}$$

$$n=3; X_{3+1} = (5 \times 85 + 2) \bmod 123$$

$$= 427 \bmod 123$$

$$X_4 = \underline{\underline{58}}$$

$$n=4; X_{4+1} = (5 \times 58 + 2) \bmod 123$$

$$= 292 \bmod 123$$

$$X_5 = \underline{\underline{46}}$$

$$\therefore \text{Sequence} = \{73, 121, 115, 85, 58, 46, \dots\}$$

- Q) generate a sequence of random no.'s (show 6 no.'s) which would be generated using LCG with modulus = 100, seed = 27, multiplier = 17 & increment = 43.

A: $X_0 = 27; a = 17; c = 43; m = 100$

$$X_1 = (17 \times 27 + 43) \bmod 100$$

$$= 502 \bmod 100$$

$$X_1 = \underline{\underline{2}}$$

$$X_2 = (17 \times 2 + 43) \bmod 100$$

$$= 77 \bmod 100$$

$$X_2 = \underline{\underline{77}}$$

$$X_3 = (17 \times 77 + 43) \bmod 100$$

$$= 1352 \bmod 100$$

$$X_3 = \underline{\underline{52}}$$

$$X_4 = (17 \times 52 + 43) \bmod 100 = 927 \bmod 100$$

$$X_4 = \underline{\underline{27}}$$

$$X_5 = (17 \times 27 + 43) \bmod 100$$

$$= 502 \bmod 100 \Rightarrow X_5 = \underline{\underline{2}}$$

$$\therefore \text{Sequence} = \{27, 2, 77, 52, 27, 2, \dots\}$$

$$\therefore \text{Period} = \underline{\underline{4}}$$

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Q)

Calculate the period for LCG with modulus = 8, seed = 4, multiplier = 5 and increment = 3.

A:

$$m = 8, X_0 = 4, a = 5, c = 3$$

$$X_0 = 4$$

$$X_1 = (5 \times 4 + 3) \bmod 8$$

$$= 23 \bmod 8$$

$$X_1 = \underline{\underline{7}}$$

$$X_2 = (5 \times 7 + 3) \bmod 8$$

$$= 38 \bmod 8$$

$$X_2 = \underline{\underline{6}}$$

$$X_3 = (5 \times 6 + 3) \bmod 8$$

$$= 33 \bmod 8$$

$$X_3 = \underline{\underline{1}} \text{ (remainder)}$$

$$X_4 = (5 \times 1 + 3) \bmod 8$$

$$= 8 \bmod 8$$

$$X_4 = \underline{\underline{0}} \text{ (remainder)}$$

$$X_5 = (5 \times 0 + 3) \bmod 8$$

$$= 3 \bmod 8$$

$$X_5 = \underline{\underline{3}}$$

$$X_6 = (5 \times 3 + 3) \bmod 8$$

$$= 18 \bmod 8$$

$$X_6 = \underline{\underline{2}}$$

$$X_7 = (5 \times 2 + 3) \bmod 8$$

$$= 13 \bmod 8$$

$$X_7 = \underline{\underline{5}}$$

$$X_8 = (5 \times 5 + 3) \bmod 8$$

$$= 28 \bmod 8$$

$$X_8 = \underline{\underline{4}}$$

$$X_9 = (5 \times 4 + 3) \bmod 8$$

$$= 23 \bmod 8$$

$$X_9 = \underline{\underline{7}}$$

$$\therefore \text{Sequence} = [4, 7, 6, 1, 0, 2, 5, 4, 7, \dots]$$

$$\text{Period} = \underline{\underline{8}}$$

Q) generate the random bit sequence (first 3 iterations) using BBSG with following parameters :-
2 primes (7 & 11) and seed = 12.

A: $p=7$; $q=11$; $X_0 = s^2 \bmod n$; $s=12$; prime = 2

$$n = p * q = 7 * 11 = \underline{77}$$

$$X_0 = 12^2 \bmod 77 = \underline{67}$$

$$X_1 = X_0^2 \bmod n = 67^2 \bmod 77 = 4489 - 4466 = \underline{23}$$

$$B_1 = X_1 \bmod \text{prime} = 23 \bmod 2 = \underline{1}$$

$$X_2 = X_1^2 \bmod n = 23^2 \bmod 77 = \underline{67}$$

$$B_2 = X_2 \bmod \text{prime} = 67 \bmod 2 = \underline{1}$$

$$\therefore \text{Sequence} = \{1, 1, 1, \dots\}$$

Q) generate the random bit sequence (first 10 iterations) using BBSG with following parameters :-
2 primes (31 & 59) and a seed of 45.

A: $p=31$; $q=59$; $s=45$; prime = 2

$$n = p * q = 31 * 59 = 1829$$

$$X_0 = s^2 \bmod n = 45^2 \bmod 1829$$

$$= 2025 \bmod 1829$$

$$= \underline{196}$$

$$X_1 = X_0^2 \bmod n = 196^2 \bmod 1829$$

$$= 38416 \bmod 1829$$

$$= \underline{7}$$

$$B_1 = X_1 \bmod \text{prime} = 7 \bmod 2 = \underline{1}$$

$$X_2 = X_1^2 \bmod n = 7^2 \bmod 1829$$

$$= \underline{49}$$

$$B_2 = X_2 \bmod \text{prime} = 49 \bmod 2 = \underline{1}$$

$$X_3 = X_2^2 \bmod n = 49^2 \bmod 1829$$

$$= \underline{572}$$

$$B_3 = X_3 \bmod \text{prime} = 572 \bmod 2 = \underline{0}$$

$$X_4 = X_3^2 \bmod n = 572^2 \bmod 1829$$

$$= \underline{1622}$$

$$B_4 = X_4 \bmod \text{prime} = 1622 \bmod 2 = \underline{0}$$

$$X_5 = X_4^2 \bmod n = 1622^2 \bmod 1829$$

$$= \underline{782}$$

$$B_5 = X_5 \bmod \text{prime} = 782 \bmod 2 = \underline{0}$$

$$X_6 = X_5^2 \bmod n = 782^2 \bmod 1829$$

$$= \underline{638}$$

$$B_6 = X_6 \bmod \text{prime} = 638 \bmod 2 = \underline{0}$$

$$X_7 = X_6^2 \bmod n = 638^2 \bmod 1829$$

$$= \underline{1006}$$

$$B_7 = X_7 \bmod \text{prime} = 1006 \bmod 2 = \underline{0}$$

$$X_8 = X_7^2 \bmod n = 1006^2 \bmod 1829$$

$$= \underline{599}$$

$$B_8 = X_8 \bmod \text{prime} = 599 \bmod 2 = \underline{1}$$

$$X_9 = X_8^2 \bmod n = 599^2 \bmod 1829 = \underline{317}$$

$$B_9 = X_9 \bmod \text{prime} = 317 \bmod 2 = \underline{1}$$

$$X_{10} = X_9^2 \bmod n = 317^2 \bmod 1829 = \underline{1723}$$

$$B_{10} = X_{10} \bmod \text{prime} = 1723 \bmod 2 = \underline{1}$$

$$\therefore \text{Sequence} = \{1, 1, 0, 0, 0, 0, 0, 1, 1, 1\}$$

Q) Calculate the period of BBS61 with following parameters:- 2 primes (103 & 211), and a seed of 100. Also display, the corresponding random bit sequence.

A: $p = 103; q = 211; \text{prime} = 2; s = 100$

$$n = p * q = 103 * 211 = 21733$$

$$X_0 = s^2 \bmod n = 100^2 \bmod 21733 = 10000$$

$$X_1 = X_0^2 \bmod n = 10000^2 \bmod 21733 = 6467$$

$$B_1 = X_1 \bmod 2 = 6467 \bmod 2 = \underline{\underline{1}}$$

$$X_2 = X_1^2 \bmod n = 6467^2 \bmod 21733 = 7797$$

$$B_2 = X_2 \bmod 2 = 7797 \bmod 2 = \underline{\underline{1}}$$

$$X_3 = X_2^2 \bmod n = 7797^2 \bmod 21733 = 6008$$

$$B_3 = X_3 \bmod 2 = 6008 \bmod 2 = \underline{\underline{0}}$$

$$X_4 = X_3^2 \bmod n = 6008^2 \bmod 21733 = 19284$$

$$B_4 = X_4 \bmod 2 = 19284 \bmod 2 = \underline{\underline{0}}$$

$$X_5 = X_4^2 \bmod n = 19284^2 \bmod 21733 = 21026$$

$$B_5 = X_5 \bmod 2 = 21026 \bmod 2 = \underline{\underline{0}}$$

$$X_6 = X_5^2 \bmod n = 21026^2 \bmod 21733 = 21723$$

$$B_6 = X_6 \bmod 2 = 21723 \bmod 2 = \underline{\underline{1}}$$

$$X_7 = X_6^2 \bmod n = 21723^2 \bmod 21733 = 100$$

$$B_7 = X_7 \bmod 2 = 100 \bmod 2 = \underline{\underline{0}}$$

$$X_8 = X_7^2 \bmod n = 100^2 \bmod 21733 = 10000$$

$$B_8 = X_8 \bmod 2 = 10000 \bmod 2 = \underline{\underline{0}}$$

$$\begin{aligned} 21026^2 \div 21733 &= 20341.99954 \\ &\approx 20341 \times 21733 \\ &= 442070753 \\ 21026^2 - 442070753 &= 21723 \end{aligned}$$

$$\therefore \text{Sequence} = \{110001001100010011000100\dots\}$$

Q) A p.t = 123 is encrypted using RSA with key generation primes 61 & 59 and public exponent = 17. Calculate the private key & cipher Text.

Ans:

$$p = 61; q = 59; M = 123$$

$$n = p * q = 61 * 59 = 3599$$

$$\phi(n) = (p-1) * (q-1) = (61-1) * (59-1) = 60 * 58 = 3480$$

$$e = 17$$

$$d = M^{-1}(e) \bmod \phi(n) = \underline{M^{-1}(17) \bmod 3480}$$

q	a	b	n	v_1	v_2	v
204	3480	17	12	0	1	-204
1	17	12	5	1	-204	205
2	12	5	2	-204	205	-614
2	5	2	1	205	-614	1433
2	2	1	0	-614	1433	-3480
	1	0		1433		

$$[v_1 - q * v_2]$$

$$\therefore \text{Private key} = 1433$$

$$C = M^e \bmod n = 123^{17} \bmod 3599 = 733$$

$$17 = 16 + 1$$

$$C \Rightarrow [(123^{16} \bmod 3599) (123 \bmod 3599)] \bmod 3599$$

$$C \Rightarrow [(123^4 \bmod 3599)^4 (123 \bmod 3599)] \bmod 3599$$

$$C \Rightarrow [((123^2 \bmod 3599)^2 \bmod 3599)^4 (123 \bmod 3599)] \bmod 3599$$

$$C \Rightarrow [((733)^2 \bmod 3599)^4 (123 \bmod 3599)] \bmod 3599$$

$$C \Rightarrow [(1038)^4 \bmod 3599] (123 \bmod 3599) \bmod 3599$$

$$C \Rightarrow (550 \times 123) \bmod 3599$$

$$C \Rightarrow 67650 \bmod 3599$$

$$C \Rightarrow \underline{\underline{2868}}$$

Q) In reference to the prev. numerical, Calculate the p.t when c.t = 2868

A: $M = C^d \bmod n = 2868^{1433} \bmod 3599$

$n = 3599$; $d = 1433$ [private key]

$\therefore d = 1433 = 2^{10} + 2^9 + 2^7 + 2^4 + 2^3 + 1$

$1433 = 1024 + 256 + 128 + 16 + 8 + 1$

$\Rightarrow 2868^{1024} \bmod 3599 = 1343$

$2868^2 \bmod 3599 = 1709$

$\Rightarrow 2868^{256} \bmod 3599 = 733$

$2868^4 \bmod 3599 = 1892$

$\Rightarrow 2868^{128} \bmod 3599 = 123$

$2868^{32} \bmod 3599 = 3173$

$\Rightarrow 2868^{16} \bmod 3599 = 2380$

$2868^{64} \bmod 3599 = 1526$

$\Rightarrow 2868^8 \bmod 3599 = 2258$

$2868^{512} \bmod 3599 = 1038$

$\Rightarrow 2868 \bmod 3599 = 2868$

$2868^{1433} \bmod 3599 = (1343 * 733 * 123 * 2380 * 2258 * 2868) \bmod 3599$

$= (2380) * (428) \bmod 3599$

$= 1018640 \bmod 3599$

$= \underline{\underline{123}}$

a — 0	j — 9	s — 18	> — 27
b — 1	k — 10	t — 19	< — 28
c — 2	l — 11	u — 20	@ — 29
d — 3	m — 12	v — 21	# — 30
e — 4	n — 13	w — 22	\$ — 31
f — 5	o — 14	x — 23	! — 32
g — 6	p — 15	y — 24	
h — 7	q — 16	z — 25	
i — 8	r — 17	? — 26	

Q) > 27

$$\begin{aligned}
 \rightarrow \text{decryption: } d &= c - \text{key} \bmod n \\
 &= 27 - 23 \bmod 33 \rightarrow (0 - 32 = 33 \text{ total set}) \\
 &= 4 \bmod 33 \\
 &= \underline{4} \rightarrow e
 \end{aligned}$$

* Additive Cipher :-

$$\begin{aligned}
 \text{encrypt: } &(c + k) \bmod n \\
 \text{decrypt: } &(c - k) \bmod n
 \end{aligned}$$

* Hill Cipher :-

$$p.t = \text{engineer} = 8/2 = 4 = 2 \times 4$$

$$= \begin{bmatrix} e & n \\ g & i \\ n & e \\ e & r \end{bmatrix} = \begin{bmatrix} 4 & 13 \\ 6 & 8 \\ 13 & 4 \\ 4 & 17 \end{bmatrix}$$

$$C = P * K \pmod{26}$$

$$= \begin{bmatrix} 4 & 13 \\ 6 & 8 \\ 13 & 4 \\ 4 & 17 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 9 \end{matrix} \pmod{26} = \begin{bmatrix} R_1 \times C_1 & R_1 \times C_2 \\ 4 \times 5 + 13 \times 3 & 4 \times 6 + 13 \times 9 \\ 6 \times 5 + 8 \times 3 & 6 \times 6 + 8 \times 9 \\ \dots & \dots \\ \dots & \dots \end{bmatrix}$$

$$= \begin{bmatrix} 59 & 141 \\ 54 & 108 \\ 77 & 114 \\ 71 & 177 \end{bmatrix} \pmod{26} = \begin{bmatrix} 59 \bmod 26 & 141 \bmod 26 \\ 54 \bmod 26 & 108 \bmod 26 \\ \dots & \dots \\ \dots & \dots \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 11 \\ 2 & 4 \\ \dots & \dots \\ \dots & \dots \end{bmatrix} = \begin{bmatrix} h & l \\ c & e \\ z & k \\ t & v \end{bmatrix} = C$$

$$D = ? = C * \det(K^{-1}) \bmod 26$$

$$K^{-1} = \begin{bmatrix} 9 & -6 \\ -3 & 5 \end{bmatrix} \Rightarrow \det(K^{-1}) = 27 \bmod 26$$

$$= MI(27) \bmod 26$$

$$\Rightarrow a = 1$$

$$\therefore D = C * K^{-1} \pmod{26}$$

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revision